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AIR FORCE MATERIEL COMMAND

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Safety

SYSTEM SAFETY GROUP INFORMATION

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This pamphlet implements AFI 91-202_AFMCSUP, *The US Air Force Mishap Prevention Program*, and describes System Safety Group (SSG) purpose, function, execution, and use. It contains a wealth of material based upon the experiences of successful and not-so-successful System Safety Groups. System Safety Groups are prescribed by AFI 91-202 and AFI 91-202_AFMCSUP, *The USAF Mishap Prevention Program*. This pamphlet applies to all system safety practitioners at the AFMC centers. This pamphlet does not apply to the Air National Guard, US Air Force Reserve, US Space Force, contractors and civilians. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with AFI 33-322, *Records Management and Information Governance Program*, and disposed in accordance with the Air Force Records Disposition Schedule located in the Air Force Records Information Management System. Refer recommended changes and questions about this publication to the Office of Primary Responsibility using the DAF Form 847, *Recommendation for Change of Publication*; route DAF Forms 847 from the field through the appropriate functional chain of command. This publication may be supplemented at any level, but all Supplements must be routed to the Office of Primary Responsibility of this publication for coordination prior to certification and approval. This AFMCPAM is for reference and guidance and is not directive in nature. Although interrelated, it does not address Nonnuclear Munitions Safety Boards per AFI 91-205, *Nonnuclear Munitions Safety Board*; Nuclear Weapon System Safety Groups per AFI 91-102, *Nuclear Weapon System Safety Studies, Operational Safety Reviews, and Safety Rules*; Directed Energy Weapons Safety Boards per AFI 91-401, *Directed Energy Systems Safety*; and Environment, Safety, and Occupational Health Working Groups.

SUMMARY OF CHANGES

Revised to reflect updated references and language from current versions of AFI 91-202 and AFI 91-202_AFMCSUP.

Chapter 1—INTRODUCTION **4**

1.1. The System Safety Program. 4

1.2. System Safety Group Description. 4

1.3. System Safety Group Functions. 4

1.4. AFMC Policy on System Safety Groups. 5

1.5. System Safety Group Members and Advisors. 6

1.6. System Safety Group Charter. 9

Chapter 2—MEETING PREPARATION **10**

2.1. Introduction. 10

2.2. Establish the Meeting Date. 10

2.3. Meeting Location. 10

2.4. Briefers. 11

2.5. Final Agenda. 11

2.6. Brief the System Safety Group Chairperson. 11

Chapter 3—THE MEETING **12**

3.1. Introduction and Welcome. 12

3.2. Review the System Safety Group Charter. 12

3.3. Review Past System Safety Group Minutes. 12

3.4. Program Status Briefing. 12

3.5. Mishap Briefing. 12

3.6. Review Safety Modifications. 12

3.7. Review Hazard Analyses. 13

3.8. Briefings. 13

3.9. Operating Problems. 13

3.10. System Safety Group Voting. 13

3.11. Concluding the Meeting: 13

Chapter 4—AFTER THE MEETING **14**

4.1. Write the Meeting Minutes. 14

4.2. Publish the Meeting Minutes. 14

4.3. Distribute the Meeting Minutes. 15

Chapter 5—SYSTEM SAFETY WORKING GROUPS 16

5.1. What is a System Safety Working Group? 16

5.2. Characteristics. 16

5.3. Examples of System Safety Working Groups. 16

Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION 17

Attachment 2—EXAMPLE: F-X SYSTEM SAFETY GROUP CHARTER 19

Attachment 3—SAMPLE BRIEFING CHARTS 23

Attachment 4—SYSTEM SAFETY GROUP ACTION ITEM 25

Attachment 5—F-X SYSTEM SAFETY GROUP QUARTERLY PROGRESS REPORT 26

Chapter 1

INTRODUCTION

1.1. The System Safety Program.

1.1.1. System safety programs, established throughout the Department of Defense, incorporate safety into equipment, systems, facilities, operations, and training. The system safety program identifies hazards and eliminates or mitigates these hazards through design selection, or controls hazards by design alterations, engineering features or devices, warning devices, and/or incorporating signage, procedures, training, and personal protective equipment.

1.1.2. System safety practitioners assigned to contractor and government program offices should:

1.1.2.1. Identify actual or potential hazards through detailed analysis of the system.

1.1.2.2. Assess each hazard in terms of worst-case mishap consequences (severity) and its expected frequency (probability).

1.1.2.3. Determine corrective actions to eliminate or mitigate hazards by design selection, or control with design alterations, engineering features and devices, warning devices, and/or signage, procedures, training, and personal protective equipment.

1.1.2.4. Verify corrective action is accomplished and produces the desired results.

1.1.3. AFI 91-202 and AFI 91-202_AFMCSUP provide the requirements for implementing effective system safety programs. Apply MIL-STD-882, *Department of Defense Standard Practice System Safety,* fundamentals to System Safety Group activities whenever possible.

1.2. System Safety Group Description. The System Safety Group includes government and defense contractor personnel associated with the particular weapon system. The System Safety Group augments the local system safety function; it is not a substitute or replacement. Therefore, many System Safety Group members are personnel not assigned to the Program Office within the command, but they advise the Program Manager or single manager on safety matters. The System Safety Group acts as an integrated product team for system safety. The System Safety Group members advise the System Safety Group chairperson on all matters pertaining to their areas of expertise and can recommend unthought-of risks, assessing these risks, and recommending solutions to these risks. Solutions can be redesign, addition of safety or warning devices, revising training or operating procedures, or accepting the risk without further actions.

1.3. System Safety Group Functions. The System Safety Group functions as an advisory group to the System Safety Group chairperson. Primarily, the System Safety Group evaluates identified hazards, recommends possible corrective actions to effectively manage the hazards until they are either eliminated or controlled. To do this, the System Safety Group has many functions which include:

1.3.1. Validating existing system safety criteria. Are current safety design criteria adequate? Is the system safety program adequate? If not, consider additional needs.

1.3.2. Establishing follow-on safety requirements. The System Safety Group members should have the necessary expertise to determine the best follow-on efforts that would apply for the system safety program.

1.3.3. Reviewing and evaluating the results of hazard analyses. Generally, the best approach is to have the results of the analyses presented to the System Safety Group.

1.3.4. Conducting independent hazard analyses. In some cases, individual System Safety Group members may have the experience, expertise, and interest to conduct an independent hazard analysis to solve a particular safety problem. Independent hazard analyses are usually accomplished outside of the System Safety Group with the results presented to the System Safety Group or as the findings become available reported to the System Safety Group members and advisors in a status report.

1.3.5. Reviewing and applying mishap lessons learned data. The System Safety Group members should have the necessary experience with the aircraft, space system, associated major subsystems, and related systems to apply lessons learned from mishaps gained from field experience.

1.3.6. Reporting safety implications of engineering and modification proposals. The program office must always perform a safety evaluation for any proposal as stated in AFI 91-202, but quite often the System Safety Group members' experience and expertise can benefit this evaluation.

1.3.7. Confirming critical failure items. The program should identify those components having a single failure that could cause a critical or catastrophic hazard or mishap and report them to the System Safety Group. Critical Safety Items identified through AFI 20-106_IP, *Management of Aviation Critical Safety Items*, should be included. Once identified the System Safety Group can provide lessons learned from other programs and recommend corrective actions, if appropriate.

1.3.8. Tracking safety modifications. The System Safety practitioner should report the status of safety modifications and safety Time Compliance Technical Orders until completion. If progress slows down or stalls, the System Safety Group could provide additional emphasis to management to get things rolling again.

1.3.9. Reporting safety implications of Category I Materiel Deficiency Reports and mishap recommendations to ensure necessary and timely action is taken for correcting safety deficiencies.

1.3.10. Providing exchange of information and experience among members.

1.3.11. Prioritizing hazards. The System Safety Group can recommend prioritizing hazards based on severity and probability so that approval authorities can better determine the optimum degree of both safety and technical risk management.

1.4. AFMC Policy on System Safety Groups. The AFI 91-202 and AFI 91-202_AFMCSUP specify the minimum requirements for System Safety Groups.

1.4.1. Most programs will have sufficient safety issues to warrant an annual meeting.

1.4.1.1. The SSG Chair can postpone the SSG, but the SSG Chair must report as stated in AFI 91-202. The SSG Chair should inform the PEO of the cause of the postponement and the estimated date when the SSG will next be held.

1.4.2. Some System Safety Groups can meet more frequently than once every 12 months due to their technical complexity if necessary.

1.4.3. Joint service acquisitions should follow the Lead Service System Safety Group augmented by the participating services.

1.4.4. SSWGs should be used to augment System Safety Groups when appropriate. The System Safety Working Group is described in **Chapter 5**. The System Safety Working Group cannot substitute for a System Safety Group.

1.4.5. The program office should send an electronic meeting announcement to all Members and Advisors not less than 45 days before the System Safety Group meeting. This announcement should give date and place of the meeting, the point of contact, and a preliminary agenda. The announcement should ask Members to provide attendee data and additional agenda items not later than 30 days before the meeting. The notice will give the primary members an opportunity to rearrange their schedules to attend the System Safety Group. An effective System Safety Group will always provide plenty of advance notice to maximize attendance.

1.4.6. The program office should provide a final agenda to each System Safety Group Members and Advisors not later than two weeks before the meeting. This will give members sufficient time to research the agenda item or to get an organization's position on the subject.

1.4.7. The program office should publish and distribute meeting minutes within 30 working days after the System Safety Group meeting to all Members and Advisors representatives whether they attended the System Safety Group or not as required by AFI 91-202.

1.4.8. As a minimum, meeting minutes should include open action items, action agencies, suspense dates, and status of proposed corrective actions. Action item status can be shown using the SSG Quarterly Progress Report in **Attachment 5.** Minutes should have enough information to describe System Safety Group deliberations and results for those that were unable to attend the meeting.

1.5. System Safety Group Members and Advisors. This paragraph describes the System Safety Group members and advisors.

1.5.1. The System Safety Group should consist of the representatives from the organizations listed in AFI 91-202 paragraph 11.4.3. Some representatives are designated as Members while others are Advisors. The System Safety Group Charter identifies which organizations have member representatives and which have advisors.

1.5.2. In general, System Safety Group members are those organizations that are required for each meeting and have the power to vote on issues that arise. Advisors, on the other hand, are those who provide guidance and advice on certain issues. They are not always required to attend each meeting and usually do not have voting power. Each program can choose its own System Safety Group members and advisors and must identify them in its System Safety Group charter as required in AFI 91-202.

1.5.3. Members. AFI 91-202, paragraph 11.4.3.1 describes the required primary Members of the System Safety Group. Additional System Safety Group Members can be added as the program requires them.

1.5.3.1. System Safety Group Chairperson. The System Safety Group Chairperson is usually the O-6 Colonel or equivalent program director, deputy program director, or Chief Engineer. In a few cases, a general officer has chaired a System Safety Group. This

provides great emphasis to a System Safety Group meeting but can stifle communications - some people are hesitant to speak their minds in front of a general officer.

1.5.3.1.1. The System Safety Group Chairperson should chair the entire meeting, not just provide introductory remarks, and then turn over the reins to someone else. The Chairperson's presence adds emphasis to the meeting. It also makes the members feel like they're really contributing to the meeting. The members will also benefit by getting the straight story directly from the Chairperson. The Chairperson benefits, too, by hearing the members' inputs firsthand. Finally, the Chairperson can make System Safety Group decisions on the spot instead of deferring them.

1.5.3.2. Program System Safety Representative. The System Safety practitioner assigned to the weapon system. This individual generally sets up including assembling the agenda and administers the meeting and writes the meeting minutes. Some programs have the weapon system contractor to deliver System Safety Group Minutes. The System Safety practitioner needs to ensure that the contract deliverable should be supplied within 30 working days after the meeting.

1.5.3.3. Lead MAJCOM Safety Office. The Lead MAJCOM Safety Office should assign a member to attend each System Safety Group. Some safety representatives may be accompanied by MAJCOM operational, maintenance, or requirements staff members.

1.5.4. Advisors. Optional, advisory representatives should be part of the System Safety Group and can be from the following organizations.

1.5.4.1. Program Engineering. Usually, members of the local engineering branch or integrated product team. They can provide technical advice and arrange for technical briefers.

1.5.4.2. Center Safety Office. The Center Safety Office will assign the full-time Center System Safety Manager, or the Center System Safety Manager will delegate System Safety Group attendance to the appropriate Center Safety Office representative as required by AFI 91-202. Due to funding issues the Center System Safety Manager will usually only physically attend those System Safety Groups that are being held at the center and require no temporary duty funds.

1.5.4.3. HQ AF Safety Center (HQ AFSEC). The flight safety officer (HQ AFSEC/SEF) or weapons safety officer (HQ AFSEC/SEW) assigned to the weapon system. Each individual serves as a primary member (AFI 91-202, Para 11.4.3.2).

1.5.4.4. HQ AFMC Safety Office. This System Safety Group advisor can come from either AFMC/SEF/SEG or SEW, AFI 91-202, Para 11.4.3.2).

1.5.4.5. Using MAJCOM. Each using command (i.e., Air Combat Command, Air Mobility Command,, etc.) should provide a member to the System Safety Group. In some cases, more than one major command should participate. For example, the F-15 System Safety Group has representatives from Air Combat Command, United States Air Forces in Europe, and Pacific Air Forces.

1.5.4.5.1. Using MAJCOM Safety Office. Each using command should provide a member from its staff safety office. Some safety representatives may be accompanied by MAJCOM operational or maintenance staff members.

1.5.4.5.2. Using MAJCOM Requirements Office. These members should ensure safety requirements are incorporated in current or future program requirements.

1.5.4.5.3. Using MAJCOM Operations Office. These members should bring operational safety issues and test issues to the System Safety Group.

1.5.4.5.4. Using MAJCOM Maintenance Office. These individuals usually attend System Safety Groups that convene in conjunction with other forums, such as product improvement working groups. These individuals can provide excellent advice for field problems, such as completing Time Compliance Technical Orders.

1.5.4.6. Numbered Air Force. The NAF flight or weapons safety officer.

1.5.4.7. Each Operational Unit (optional). The Flight Safety Officer from each operational unit (wing or group) possessing the weapon system or, in the case of missiles, the missile safety officer. For some System Safety Groups, this can be many personnel. There have been over 100 attendees at System Safety Group meetings. In this case, it may be better to have the MAJCOM safety office represent the individual operational units.

1.5.4.8. Air Force Sustainment Center. Air Force Sustainment Center may send a representative to either discuss or report programmed depot management issues or supply chain issues relevant to the System Safety Group.

1.5.4.9. Foreign Military Sales Programs. Foreign Military Sales Programs will result in System Safety Group members from other countries possessing the weapon system. This could create additional problems. Sanitize mishap data for presentation at System Safety Group meetings, so there will be no releases of privileged safety information (DAFI 91-204, *Safety Investigations and Hazard Reporting*, Chapter 4). You may have an awkward situation if two countries possess the same weapon system but are not friendly to each other. Holding System Safety Groups off base may alleviate some base security concerns.

1.5.4.10. Test Centers. The Air Force Test Center should be an advisor when involved with the weapon system. Lead Developmental Test and Evaluation Organization and Participating Test Organizations should also be invited. NOTE: The 96 Test Wing (TW)/SES personnel will greatly assist in determining if the weapon system can be approved by the Non-Nuclear Munitions Safety Board. 96 TW/SES Office is the engineering/technical advisors to the Non-Nuclear Munitions Safety Board and its expertise is recognized by the Non-Nuclear Munitions Safety Board and the Navy's Weapon Explosive Safety and Evaluation Review Board. 96 TW/SES' review of the weapon system design, modification, and/or integration onto the aircraft will pay big dividends in the long run. The local flight test office can inform the System Safety Group on the flight tests of modified aircraft.

1.5.4.11. Supporting Program Offices. Other program offices may send personnel to support the System Safety Group as required. For example, AFLCMC/LP, Tinker AFB, may send an engine representative.

1.5.4.12. Contractors. Airframe and major subsystem (e.g., engine) contractors may provide technical advisors to System Safety Group meetings. Having the contractors present will provide support in defining safety hazards and outlining options to mitigate the risk. If contractors will be at the System Safety Group, the program office contracts

branch will need to be informed to ensure that issues and possible mitigation ideas discuss at the System Safety Group will not be interpreted as direction to the contractor and possibly expanding the scope of an existing contract.

1.5.4.13. Human Systems Integration. Human Systems Integration analysis and assessments focus on the role of the human in the Department of Defense acquisition process are critical to weapons system development. Therefore, the Center Human Systems Integration should be invited participate in the System Safety Group.

1.5.4.14. HQ Air Force. Some Secretary of the Air Force (SAF) offices could attend high visible AF System Safety Groups to keep abreast of safety issues. SAF/AQ usually attends high visibility developing aircraft program System Safety Groups, while SAF/IEE or AF/TE may send attendees to other System Safety Groups.

1.6. System Safety Group Charter. The System Safety Group is a chartered group. The charter will establish the purpose and scope of a System Safety Group, its function, limitations, and membership. It should include.

1.6.1. Outline the System Safety Group tasks and responsibilities and provide the authority for assigning and completing action items.

1.6.2. Cover the procedures for handling System Safety Group recommendations and action items including disagreements.

1.6.3. List member organizations, particularly those responsible for setting up and chairing the meeting. The list should consist of member organizations, not individual names, which change frequently.

1.6.4. Coordinate the draft charter with the proposed System Safety Group Members and Advisors. A sample System Safety Group charter format is in **Attachment 2.**

1.6.5. Members should approve the charter at the first meeting or shortly thereafter.

1.6.6. The final charter should be signed by the System Safety Group Chairperson.

1.6.7. The System Safety Group Charter should be reviewed by the System Safety practitioner each year 60 days before the System Safety Group and provide updates to the System Safety Group Members with the draft System Safety Group agenda.

Chapter 2

MEETING PREPARATION

2.1. Introduction. Coordination on the draft System Safety Group charter has been received and it is time to hold the first meeting. This chapter describes the final preparations for the first and subsequent meetings. The chapter is written to describe the efforts for a separate System Safety Group. Many organizations prefer to combine a System Safety Group with another meeting, such as a program management review or product improvement working groups meeting. There is nothing wrong with this idea, and it provides a chance to save considerable man-hours and temporary duty costs.

2.2. Establish the Meeting Date. The System Safety Group Action Officer should:

2.2.1. Be sure the System Safety Group Chairperson and the System Safety practitioner will be available. These are the key people. The System Safety Group Chairperson chairs the meeting, but the System Safety practitioner really runs things. It's a very good idea to coordinate the proposed date with the local Center System Safety Manager and the AFMC system safety office (HQ AFMC/SES) to be sure there are no conflicts with other meetings. After confirming the date, the System Safety practitioner should send notification to all System Safety Group members at least 45-60 days before the meeting date. In this notification, ask for attendee data (name, grade, organization, etc.) and ask for new agenda items. Be sure to get this information at least 30 days before the meeting so you can send out a final agenda two weeks before the meeting.

2.2.2. Ask for detailed agenda items. Some System Safety Group members will provide only a list of subjects. They should provide background information and state expectations of the System Safety Group (i.e., design fix, special procedures, etc.). If a particular member fails to provide adequate details, follow up with the member and get the details. If you ignore the item due to insufficient knowledge, be prepared for the item to come up at the meeting, anyway, because it probably will. This has been seen before with the Program Manager and their staff unprepared to address the issue, because they were not notified beforehand.

2.2.3. Be ready to change the meeting date if a schedule conflict occurs.

2.3. Meeting Location. Set the place for the meeting. Obviously, this can affect the meeting date: if the favorite conference room isn't available on a particular meeting date, consider moving it to another date. About the meeting location:

2.3.1. Avoid auditoriums: use a conference room setup. Have the System Safety Group members feel like meeting participants, not observers.

2.3.2. Anticipate a crowd. Some members bring additional people. In addition, many unannounced people will show up. It's tough to accurately guess the number of attendees. System Safety Groups have seen as few as six and as many as over 100 attendees.

2.3.3. Use a centrally located conference room, such as the organization’s main conference room. This will allow easy access to engineers and technicians that may have answers to problems being worked by the System Safety Group. The downside is that higher priority meetings may bump System Safety Group out of the room, so having a back-up plan would be advisable.

2.3.4. Sometimes contractor facilities host System Safety Groups instead of program offices. This has the advantages of tapping the experts who are on-site to answer questions and showing the production line or actual hardware. The only disadvantage is the administrative headache of visitor clearances to the facility.

2.3.5. Some program offices have held a System Safety Group via Microsoft Teams or Zoom.gov (i.e., Virtual). This can save a bundle of temporary duty costs, but there are other factors or issues to consider:

2.3.5.1. Although more people are getting used to virtual meetings, there are still a few of people who are hesitant to speak up while 'on the air'. Recommend that virtual meetings be used whenever practical. People are getting more and more used to the idea and are more amenable to using a virtual meeting to save valuable temporary duty costs. For small groups, such as System Safety Working Groups, a virtual meeting can be very practical.

2.4. Briefers. Give advance notice to the folks that will be briefing the System Safety Group. Schedule the briefers according to the System Safety Group agenda. Be sure to get the right person to do the briefing. Sample briefing chart examples can be seen in **Attachment 3.** A substitute briefer may not be able to answer all the questions.

2.5. Final Agenda. Provide the final agenda to each System Safety Group member at least two weeks before the meeting. The agenda should list the subject, times, and briefer or Office of Primary Responsibility. Some System Safety Groups provide an agenda package that gives background information and points of contact for each agenda item. Such a package allows each System Safety Group member to be better prepared for each agenda item. This should help avoid any surprises or misunderstandings at the meeting.

2.6. Brief the System Safety Group Chairperson. It's important to brief the System Safety Group Chairperson as to what to expect from the System Safety Group before the System Safety Group. This will assure that Chairperson is familiar with all System Safety Group items and will be ready for controversial items or other possible surprises. The System Safety practitioner should sit down with the System Safety Group Chairperson and go through the agenda.

Chapter 3

THE MEETING

3.1. Introduction and Welcome. The System Safety practitioner calls the meeting to order, introduces themselves, and welcomes the attendees, goes over the ground rules, and asks the System Safety Group chairperson for opening comments. The System Safety practitioner then provides any administrative remarks, and each System Safety Group member and attendees should introduce themselves to the group. This will ease communications and give members a chance to know each other better.

3.1.1. During administrative remarks the System Safety practitioner or facility host should describe where the restroom and refreshment facilities are located in relation to the meeting room and what to do in an emergency (tornado, fire, etc.).

3.1.2. Recommend that all cell phones and other electronic devises be turned off or, at least, turned to vibrate during the meeting. Nothing is more annoying than a cell phone ringing during a meeting. Some facilities block electronic signals while inside. If that is the case, attendees need to be informed so they can step outside during breaks to check messages.

3.2. Review the System Safety Group Charter. Review it for accuracy and currency. Usually, the only changes are to the membership list. The best approach for reviewing the charter is to pass/send out copies to the members when the final agenda is sent out and have them recommend any changes, preferably before the System Safety Group. Don't read aloud the entire charter; that can get boring.

3.3. Review Past System Safety Group Minutes. Ask the System Safety Group members if there are any recommended changes to the minutes of the last meeting. Normally, there aren't any, but ensure clarification of errors or omissions.

3.4. Program Status Briefing. After the introductions, the System Safety Group chairperson should give a briefing on the status of the weapon system program. This briefing should cover the fleet status, major milestones, and significant modifications and other efforts. Some System Safety Group chairpersons use a briefing that was prepared for a program assessment review or other purpose. In any case, the briefing will set the stage for the System Safety Group.

3.5. Mishap Briefing. The System Safety Group member from HQ AFSEC should give a mishap briefing that shows mishap rates, trends, high-lights, and significant mishaps since the last meeting. AFSEC does not always prepare the mishap briefing charts. Whoever prepares the mishap briefing charts, be sure to sanitize these briefings, i.e., omit specific names, places, and tail numbers, so that the System Safety Group doesn’t run into the privileged information problem (DAFI 91-204, Chapter 4). This briefing should also include the AFSEC areas of high interest and concern.

3.6. Review Safety Modifications. System Safety Groups review the status of safety modifications at each meeting. The purpose of this review is to have System Safety Group members identify problems with getting a safety modification completed and to recommend corrective action. This review also gives the System Safety Group members an opportunity to see what safety modifications are being done. Some System Safety Groups have varied their definition of safety modifications. The following has been included in different System Safety Groups:

3.6.1. Safety modifications.

3.6.2. Safety-coded Engineering Change Proposals.

3.6.3. Safety-coded Time Compliance Technical Orders.

3.6.4. Recommendations from significant (e.g., Class A) mishaps.

3.6.5. Someone's own idea of "what affects safety".

3.6.6. Review of the mishap messages with AFMC recommendations that have occurred since the last meeting.

3.6.7. Which of these require review? All of them; they all affect the overall safety of the weapon system. How is the review done? Review can be done by any suitable method, such as handouts or briefing charts listing the modifications and their status. A knowledgeable person should be there to answer questions.

3.7. Review Hazard Analyses. Occasionally, a System Safety Group will need to review a hazard analysis. If this should happen, arrange for a presentation of the results of the analysis, i.e., significant hazards identified and their corrective action. Some System Safety Group members might lack the specific training to conduct a detailed review of a hazard analysis, but all the members take interest in the results. There may be a few members who wish to review a detailed analysis. These members can form a separate System Safety Working Group to review the analysis and report back to the System Safety Group.

3.8. Briefings. A briefing of some sort covers most System Safety Group discussion and action items. Some briefings are provided by the briefer without visual aids, but the majority use charts or other visual aids. Based on other System Safety Groups, here are some lessons learned (constructive criticism) about briefings.

3.9. Operating Problems. During the meeting, several operating problems may crop up. The following subparagraphs describe potential problems and methods of coping with them. Most of the responsibility to run a smooth System Safety Group lies with the Chairperson. Usually, the Chairperson delegates the actual System Safety Group operation to the System Safety practitioner, so the System Safety practitioner will need a good understanding of System Safety Group philosophy and functions.

3.10. System Safety Group Voting. Depending on the System Safety Group Charter, the Chairperson should ask for the System Safety Group Members to vote on issues: e.g., closing previous System Safety Group action items, accepting new action items, etc.

3.11. Concluding the Meeting: A good meeting needs a good conclusion. Summarize key items and read aloud (or pass out) all action items so that everyone goes home with a complete understanding of each task. This final action item review should identify the specific action, who should do it, and when. Identify specific suspense dates.

Chapter 4

AFTER THE MEETING

4.1. Write the Meeting Minutes. There is no standard or regulatory format for minutes of the meeting. As a result, some minutes are in letter format, while others are a package or book with each page detailing an action item. The format is not directed by AFMC, as long as the minutes communicate essential information i.e., what happened during the meeting? Minutes should include at least the following:

4.1.1. The purpose of the System Safety Group Minutes (determined by the program office).

4.1.1.1. AFMC recommends the System Safety Group Minutes can be used to document the Program Manager's acceptance of low and medium risks.

4.1.2. The agenda that can serve as a "table of contents."

4.1.3. The System Safety Group charter so members should have the most current version. A link to an electronic System Safety Group charter is acceptable.

4.1.4. The list of attendees with organizations and telephone numbers.

4.1.5. Narrative summary of the System Safety Group proceedings for each discussion item and action item. The minutes don't need to be long, but there should be enough narrative so that members who missed the meeting will know what happened. For reference purposes, the minutes should include the rationale for all System Safety Group decisions.

4.1.5.1. The System Safety Group minutes should not be a word-for-word transcript of the meeting, nor should the minutes be overly sanitized.

4.1.6. Presentation materials, such as briefing charts, are optional. Usually, the best approach is to provide charts that really help the minutes. Some System Safety Group minutes include no charts while others include every single chart. It's up to the individual System Safety Group. If the minutes include many charts, reduce the size by printing two charts on a single page, unless the charts cannot be read in the reduced size.

4.1.7. A list of action items. An example of action item content can be seen in **Attachment 4.** This list should spell out what tasks, who will do it, and when. Be sure the action agency has accepted the action item; don't surprise them in the minutes.

4.1.8. Plans for the next meeting. Usually, the exact date and location will not be available, but at least include the general location and approximate date (e.g., 'June 2023 at Wright-Patterson AFB').

4.2. Publish the Meeting Minutes. Sometimes publishing the minutes can be tougher than writing them. Minutes need to be written, then approved by the System Safety Group chairperson. They can be copied and then distributed. The total process takes some time which can vary greatly. (Some programs strive for as few as eight days between the meeting and distribution of minutes and some has taken 5 months. AFMC policy is 30 days after the meeting, and all System Safety Groups should be able to meet that deadline.) Two easy ways to distribute minutes: e-mail, and the Internet. (In the case of the internet, publish the minutes on your organization's home web page.) Both methods can be significant time and cost savers, and nearly everyone has the technical

capability, so this can work very well. Supporting briefing charts can easily accompany the minutes in either case.

4.3. Distribute the Meeting Minutes. Recommend that System Safety Group meeting minutes be distributed to the following:

4.3.1. Each System Safety Group member, whether they attended or not.

4.3.2. Each office that provided an attendee, even if that office is not a member.

4.3.3. The HQ AFMC System Safety office.

4.3.4. The Air Force Safety Center.

4.3.5. Your Center safety office.

Chapter 5

SYSTEM SAFETY WORKING GROUPS

5.1. What is a System Safety Working Group? Think of a System Safety Working Group as an informal, local subset of a System Safety Group. A System Safety Working Group is usually formed when a full System Safety Group wants to research a problem but does not want to tie up the full membership. The System Safety Group will generally form a System Safety Working Group to work a problem separately and report back to the System Safety Group. A System Safety Working Group augments a System Safety Group; it's not a substitute for a System Safety Group.

5.2. Characteristics. A System Safety Working Group operates somewhat like a System Safety Group but has different characteristics.

5.2.1. A System Safety Working Group usually works a single problem or issue, then disband - much like a process action team.

5.2.2. System Safety Working Group membership is limited, usually to local engineering and safety personnel, and to other interested parties. Membership depends on the given situation. There are no regulatory requirements for System Safety Working Group members.

5.2.3. System Safety Working Groups generally don't need a charter. The overarching System Safety Group charter would cover the System Safety Working Group.

5.2.4. The local System Safety practitioner is a more appropriate chairperson for System Safety Working Groups, not the System Safety Group Chairperson.

5.2.5. There are no requirements for System Safety Working Group meeting minutes; however, the System Safety Working Group may need to write minutes to document their findings and present them to the System Safety Group and/or the USAF Airworthiness Board.

5.3. Examples of System Safety Working Groups. System Safety Working Groups can be formed for many problems. Examples are:

5.3.1. Focused evaluation of selected hazards that need to be accepted by the proper approving authority before the next program milestone (Preliminary Design Review, Critical Design Review, First Flight, etc.).

5.3.2. Evaluating the system safety program for an engine used in an aircraft.

5.3.3. Assisting a contractor performing a hazard analysis on modified weapons jettison circuit.

5.3.4. Recommending employment of a new fire detection and fire suppression system.

5.3.5. Drafting aircrew procedures during hot refueling of an aircraft.

DANNY A. SLIFER, Colonel, USAF
Director of Safety

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 20-106_IP, *Management of Aviation Critical Safety Items*, 25 Jan 2006

AFI 91-102, *Nuclear Weapon System Safety Studies, Operational Safety Reviews and Safety Rules*, 22 May 2019

AFI 91-202, *The US Air Force Mishap Prevention Program*, 12 Mar 2020

AFI 91-202_AFMCSUP, *The US Air Force Mishap Prevention Program*, 22 Mar 2022

AFI 91-205, *Nonnuclear Munitions Safety Board*, 23 May 2018

AFI 91-401, *Directed Energy Systems Safety*, 28 Nov 2018

DAFI 91-204, *Safety Investigations and Hazard Reporting*, 10 Mar 2021

MIL-STD-882E, *DoD Standard Practice, System Safety*, 11 May 2012

Adopted Forms

DAF Form 847, *Recommendation for Change of Publication*

Abbreviations and Acronyms

AFSEC—Air Force Safety Center

SSG—System Safety Group

TW—Test Wing

Office Symbols

96 TW/SES—96 Test Wing, System Safety Division

AF/TE—Headquarters Air Force/Test and Evaluation

AFLCMC/LP—Air Force Life Cycle Management Center, Propulsion Directorate

HQ AFMC/SES—Headquarters Air Force Materiel Command, System Safety Division

HQ AFMC/SEF—Headquarters Air Force Materiel Command, Flight Safety Division

HQ AFMC/SEG—Headquarters Air Force Materiel Command, Ground Safety Division

HQ AFMC/SEW—Headquarters Air Force Materiel Command, Weapons Safety Division

HQ AFSEC—Headquarters Air Force Safety Center

HQ AFSEC/SEF—Headquarters Air Force Safety Center, Flight Safety Division

HQ AFSEC/SEW—Headquarters Air Force Safety Center, Weapons Safety Division

SAF/AQ—Secretary of the Air Force/Acquisition

SAF/IEE—Secretary of the Air Force/Environment, Safety and Infrastructure

Terms

Evaluation—Method of appraising the effectiveness of system safety program management. Addresses the areas of commander supervisory support, compliance with program directives and the effectiveness of mishap prevention programs (performance).

Full-time—Individuals in primary duty system safety positions. See definition for safety and health officials in AFI 91-202.

Hazard—A condition, procedure or practice that creates a potential for producing death, injury, illness, fire, property damage, equipment damage or environmental damage.

Major Command—For the purpose of system safety, MAJCOM includes Air Combat Command, Air Education Training Command, Air Force Materiel Command, Air Force Reserve Command, Air Force Global Strike Command, Air Force Space Command, Air Force Special Operations Command, Air Mobility Command, Air National Guard, Pacific Air Forces, and United States Air Forces in Europe.

Major Subsystem—A subsystem to the aircraft without which the mission of the platform could not be accomplished successfully. Major subsystems include, but are not limited to, propulsion, ejection system, radar systems.

Mishap—A mishap is an unplanned occurrence, or series of occurrences, that results in damage or injury and meets Class A, B, C, D and E mishap reporting criteria IAW DAFI 91-204.

Risk Management—The application of a systematic process or thinking to detect, assess, and control risk to enhance total organizational performance.

Safety—The programs, RM activities, and organizational and cultural values dedicated to preventing injuries and accidental loss of human and material resources, and to protecting the environment from the damaging effects of DoD mishaps.

System Safety Groups (SSGs)—Augment the program office system safety function; it is not a substitute or replacement. While many SSG members are not assigned to the Program Office, they advise the system program manager or single manager on safety matters. They act as an integrated product team for system safety. The members assist the program office in identifying risks, assessing these risks, and recommending solutions to these risks. The SSG includes safety experts associated with the weapon system.

System Safety Working Groups (SSWGs)—Are a subset of SSGs. SSWGs are usually formed when a full SSG wants to research a problem without tying-up the full membership. The SSG will generally form an SSWG to work a problem separately and report back to the SSG. An SSWG augments an SSG; it's not a substitute.

Attachment 2

EXAMPLE: F-X SYSTEM SAFETY GROUP CHARTER

A2.1. Introduction: Following is an example for a System Safety Group Charter. Contact AFMC Center Safety Office for the Center System Safety Group Charter template.

A2.1.1. A System Safety Group is required in accordance with AFI 91-202, Chapter 11 and AFI 91-202_AFMCSUP. A System Safety Working Group may also be established and convened at the direction of the System Safety Group Chairperson. The System Safety Group membership and operating procedures detailed herein are intended to fulfill the applicable requirements of Air Force policy directives established to ensure the safety of the F-X weapon system operations.

A2.2. Purpose:

A2.2.1. The F-X System Safety Group is established to consider actual and potential safety problems associated with operation and maintenance of the F-X weapon system with emphasis on mishap prevention. The System Safety Group is to identify, discuss, and define safety issues related to the F-X weapon system. The purpose is to provide the Program Manager risk assessments with recommendations on the relative priorities when resolving valid safety deficiencies/issues. Decisions of the System Safety Group will constitute recommendations to the System Program Director. The System Safety Group or System Program Director may withhold a decision until a Program Management Review or Executive Review, as appropriate.

A2.3. Functions:

A2.3.1. Coordinate all safety related activities pertaining to the F-X weapon system.

A2.3.2. Take advantage of special experience and expertise of members and advisors.

A2.3.3. Identify, track, and resolve specific safety problems or deficiencies.

A2.3.4. Review, evaluate, and augment the safety analyses of hazards.

A2.3.5. Consider differing viewpoints on the systems' design, operation, or test results before arriving at a safety position.

A2.3.6. Identify safety critical items and assess critical single-point failures.

A2.3.7. Review safety related engineering change proposals, deviations, and waivers.

A2.3.8. Consider lessons learned on similar aircraft and missions to help avoid or eliminate potential hazards.

A2.3.9. Monitor corrective action on safety deficiencies which have been identified by Hazard Analyses, mishap investigations, and deficiency reports.

A2.3.10. Review testing procedures that are brought before the group which are used to verify safety of flight requirements.

A2.3.11. Ensure Technical and Flight Manuals (dash one) are thoroughly reviewed to determine that necessary caution and warning notes and emergency procedures are included and update as necessary.

A2.3.11.1. Act as a sounding board for safety issues identified by its members.

A2.3.12. Review aircraft modifications for safety considerations.

A2.4. Authority:

A2.4.1. AFI 91-202, and AFI 91-202_AFMCSUP, provides the authority to establish the System Safety Group. All findings of the System Safety Group constitute recommendations to the System Program Director.

A2.5. Chairperson:

A2.5.1. The Program Manager, Deputy Program Manager, or Chief Engineer is the Chairperson of the System Safety Group. Action items should be assigned by the System Safety Group Chairperson, who has final approval authority for the System Safety Group minutes.

A2.6. Membership:

A2.6.1. Permanent memberships for the System Safety Group should include representatives from the organizations listed in the attachment to this charter. Members may be added or deleted from the list as deemed necessary by the Chairperson.

A2.6.2. Members are selected because of their close association with the operation, maintenance, design, testing, and deployment of the system. Member participation in the System Safety Group is essential to provide an understanding and interpretation of the requirements from which the system design developed, and to provide an input for their organizations on issues that arise throughout the life cycle of the F-X weapon system.

A2.6.3. Membership responsibilities:

A2.6.3.1. Respond to action items assigned by the Chairperson.

A2.6.3.2. Have the authority to express or be responsible for getting the official position of their organization regarding the F-X system safety issues.

A2.6.3.3. When notified of a System Safety Group meeting, submit to the Program Manager recommended agenda items as required. Be prepared to brief the System Safety Group on the proposed agenda item, describing the area of concern.

A2.6.3.4. When notified of the System Safety Group agenda items, be prepared to discuss, and act upon the agenda items.

A2.6.3.5. When assigned action items, be prepared to report the findings and results by the established suspense to the System Safety Group.

A2.6.3.6. Attend all F-X System Safety Group meetings. If an alternate is sent, that person should be thoroughly briefed on the status of the program and be prepared to actively participate in the meeting.

A2.6.3.7. Be familiar with the applicable system safety documents.

A2.7. Advisors:

A2.7.1. Advisors to the System Safety Group should include, but not be limited to, representatives from organizations listed in the attachment. Additional advisors may be summoned as deemed necessary by the Chairperson.

A2.7.2. Advisors provide specialized expertise to the Chairperson and the System Safety Group with updates on regulatory and policy changes as they pertain to the F-X program.

A2.7.3. Advisor responsibilities:

A2.7.3.1. When assigned action items, be prepared to report the findings and results by the established suspense to the System Safety Group.

A2.7.3.2. Thoroughly study issues assigned to them and provide a qualified opinion and/or recommended action, as applicable, to the System Safety Group.

A2.7.3.3. Act as a safety liaison between their respective organization and the F-X System Safety Group.

A2.7.3.4. Furnish their organization's policy guidance applicable to the F-X System Safety Program.

A2.7.3.5. Provide additional, qualified guidance and information to the System Safety Group on F-X safety issues as related to their specialty.

A2.8. Administration:

A2.8.1. The central point of contact and coordination for System Safety Group meetings is ABC/XYZ, Building 350, 750 3rd Street, Smythington AFB CA.

A2.8.2. The system safety practitioner should record System Safety Group meeting proceedings and generate and distribute meeting minutes.

A2.8.3. The System Safety practitioner should ensure approved meeting minutes are compiled and distributed to all System Safety Group members, advisors, and attendees within 30 days following the meeting.

A2.8.4. Report on action items should be in briefing format to the System Program Director and included in the System Safety Group minutes.

A2.8.5. Recommendations should be annotated in the minutes.

A2.8.6. All organizations listed in the attachment are responsible for providing to the System Safety practitioner a list of names designating their primary and alternate members of the System Safety Group.

A2.8.7. The basic approach for the F-X System Safety Group should be to require majority agreement. However, the final decision lies with the System Safety Group chairperson. In the case of an unresolved disagreement, individual reports may be submitted to the System Safety Group chairperson within 10 calendar days after the System Safety Group meeting for inclusion into the minutes as a matter of record. Unresolved issues may become agenda items for the Program Management Review or Executive Review.

A2.8.8. Changes to the charter can be recommended to the System Safety Group by any member. Changes to the charter should be discussed along with the recommended written change submitted to the System Safety Group Chairperson. This should be annotated in the minutes and discussed and voted on at the next System Safety Group. A simple majority should rule.

A2.9. System Safety Working Group:

A2.9.1. A System Safety Working Group can be established at any time by the System Safety Group Chairperson for the purpose of evaluating and providing recommendations to the System Safety Group on specific safety issues. The System Safety Working Group would therefore be a subset of the System Safety Group. Membership of the System Safety Working Group should consist of selected System Safety Group members plus selected technical advisors as deemed necessary by the System Safety Group Chairperson. The findings of the System Safety Working Group should be reported to the System Safety Group Chairperson and reviewed at the subsequent System Safety Group.

A2.10. General:

A2.10.1. Action Item Log. The Action Item log should be reviewed by all members on an "as required" basis. If a disagreement in closure status exists, or if a significant level of risk (High or Serious) should be accepted as required by AFI 91-202, the issue will be addressed by the System Safety Group. The Action Item log status should be presented at each System Safety Group meeting. All Action Items and Hazards should be tracked. The database should be updated by the System Safety practitioner.

A2.10.2. The System Safety Group should meet at least twice a year, approximately 45 days prior to the Program Management Review and Executive Review. The schedule should be determined by the Program Manager.

A2.10.3. Notice of System Safety Group meetings should be forwarded to members and advisors at least 45 days before the meeting.

A2.10.4. The System Program Director will make the final decision on all F-X safety issues. Consideration will be given to the inputs and recommendations of System Safety Group members and advisors.

A2.11. Attachment To Charter.

A2.12. System Safety Group Members (AFI 91-202, 11.4.3).

A2.12.1. Program Manager /Program Manager Deputy/Chief Engineer (Chair)

A2.12.2. Program System Safety Manager

A2.12.3. HQ AFSPC/SE (for space)

A2.12.4. AFMC/SES System Safety Office

A2.12.5. Lead MAJCOM Safety Office

A2.12.6. Lead Operational and Developmental Test Agencies

A2.13. System Safety Group Advisors.

A2.13.1. Center System Safety Manager

A2.13.2. AFSEC/SE (for non-space systems other than ACAT 1)

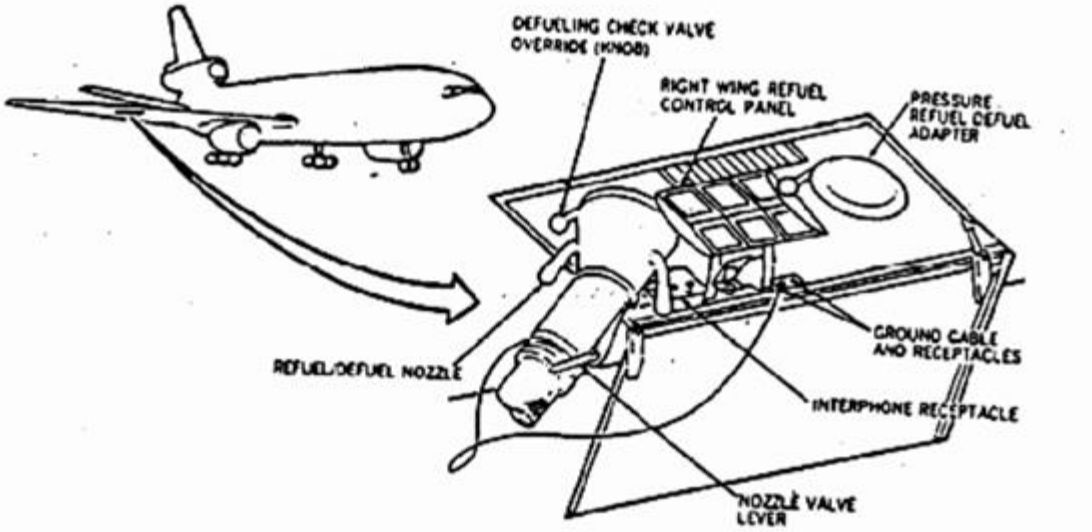
A2.13.3. AFMC/SE

Attachment 3

SAMPLE BRIEFING CHARTS

A3.1. Briefing Chart Example. Following are examples for possible System Safety Group briefing charts. Contact AFMC Center Safety Office for the Center System Safety Group briefing chart template.

Figure A3.1. Sample Briefing Chart.



SSG ITEM: 51-2

HAZARD: REFUELING POPPET FAILS TO SEAT UPON DISCONNECT

EFFECT: FUEL LEAK FROM RECEPTACLE (RHI 12)

EXPERIENCE: ONE TO DATE (MANUFACTURING TOLERANCES)

MITIGATION: RE-DESIGN POPPET SPRING
INSTALL SHUTOFF VALVE
RE-ATTACH NOZZLE (INTERIM MEASURE)

STATUS:

REVISED RHI:

Figure A3.2. Sample Quad Briefing Chart.

U.S. AIR FORCE

SAFETY ACTION ITEM: 14-007

Inadvertent PORC Impingement

AFLCMC… Providing the Warfighter's Edge

RAC	882A	882E	Level
Initial	I/8	1C	HIGH
Mitigated	I/12	1D	SERIOUS
Final	Eliminated by Design		
Acceptance	PEO		

HAZARD DESCRIPTION:

Primary Oscillating Refueling Cog (PORC) requires continuous re-torqueing during ground refueling operations, resulting in MX personnel injury/death from sprocket/cog impingement.



ACTION ITEM SOURCE:

Class A Mishap;
AFSAS Recommendation # 1037761

OPTIONS TO RESOLVE / MITIGATE HAZARD:

1) Reduce PORC Oscillations during refueling; TCTO 7Q-3987
   - Slowed from 30 to 10 Cycles/Minute
   - Dash-6 Warning Added
2) Field Design Modification; TCTO 8J-F19-543
   - Locking PORC Nuts eliminate need for re-torque

HAZARD STATUS

1)	Risk assessment completed	11 Oct 2013
2)	TCTO 7Q-3987 Issued	28 Oct 2013
3)	PEO Accepted Serious Risk	22 Mar 2014
4)	Modification Analysis Complete; Contract with YoYoDyne	1 Apr 2014
5)	TCTO 8J-F19-543 Issued	29 Jun 2014
	• 5 of 87 Kits Fielded/installed	8 Aug 2014
	• Installation ECD	3 Jan 2017

3

Attachment 4

SYSTEM SAFETY GROUP ACTION ITEM

A4.1. Action Item Example. Following is an example for possible System Safety Group Action Item. Contact AFMC Center Safety Office for the Center System Safety Group Action Item template.

Table A4.1. Suggested System Safety Group Action Item Format.

ITEM NO:	
DATE:	
INITIATOR:	
SUBJECT:	
BACKGROUND:	
RISK HAZARD INDEX/ RISK ASSESSMENT CODE:	
DISCUSSION:	
RECOMMENDED ACTION:	
ACTION AGENCY:	
SYSTEM SAFETY GROUP ACTION:	
STATUS:	
REVISED RISK HAZARD INDEX/RISK ASSESSMENT CODE:	
SIGNATURE:	
DATE:	

Attachment 5

F-X SYSTEM SAFETY GROUP QUARTERLY PROGRESS REPORT

A5.1. Progress Report Example. Following is an example for possible System Safety Group Progress Report. Contact AFMC Center Safety Office for the Center System Safety Group Progress Report template.

Table A5.1. Suggested System Safety Group Progress Report.

ITEM NAME:
REPORT PERIOD: ACTION OPR:
PROBLEM SUMMARY AND SOLUTION/PROPOSAL: PROGRESS DURING REPORT PERIOD: APPROXIMATE PERCENT COMPLETE:
CURRENT COMPLETION SCHEDULE/MILESTONES: PLANS FOR NEXT 90 DAYS: REMARKS/ROADBLOCKS/SUGGESTIONS:
REPORT PREPARED BY:
APPROVED BY:

Table A5.2. Percent Complete Criteria.

5% - Problem assigned	55% - Prototype completed
10% - Exhibit(s) requested	60% - Prototype proven
15% - Exhibit(s) received	65% - Design work: completed
20% - Analysis underway	70% - Procurable package prepared
25% - Analysis completed	75% - TCTO proposal to CCB
30% - Solution(s) proposed	80% - TCTO approved by CCB
35% - Solution/action selected	
40% - Schedule/costs obtained/approved	90% - All documentation completed
45% - Purchase description available	95% - Installation started
50% - Contract authorized	100% - Item closed/completed